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Correlation coefficient distribution and its application in the comparison of chemical data sets

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ABSTRACT

This paper presents a comprehensive statistical framework in order to rigorously derive the correlation coefficient distribution between two independent chemical data sets. The novel approach not only allows for unequivocal interpretation of distribution parameters and confidence intervals but also the generalization of previous results for fixed size comparisons into the new and experimentally important case of variable dimension data assessments. Original theoretical results include an objective criterion ("number of nines") for the coefficient of multiple determination in mathematical model fitting (e.g. calibration curve acceptance in validation protocols) and the requirement of a minimum correlation coefficient match of approximately 70% in variable size evaluations. Significance tests have been applied to the drug profile linkage of 532 real case cocaine samples seized by the National Police of Peru between the years of 2021 and 2022.

1. Introduction

The statistical concept of correlation which was originally proposed by Galton [1], has always been an important topic in all areas of science. Pearson [2] first explicitly derived the index to measure bivariate association (the product-moment mean centered correlation coefficient, $\boldsymbol{r}$). It was the first dimensionless, normalized $[-1; 1]$ and linear transformation invariant correlation measure which is still widely used in several fields such as experimental design, mathematical modeling, trend analysis etc.

The bivariate case is commonly thought of as representing observations of two different characters on the same sample [3]. Its concept has been generalized to include the multiple correlation between one variate and a linear combination of variables in a group of common measurements. Fisher was the first one to derive the sampling bivariate [4] and multiple correlation [5] coefficient distributions based upon geometrical arguments and normality assumptions. Alternative expressions have been formulated for the cases of null and nonzero population correlation coefficients [3,6]. More recently, Pecka and Ponec [7] used spherical coordinates to derive analytical expressions for the Pearson coefficient cumulative distribution in order to compare competing Quantitative Structure–Activity Relationship (QSAR) models.

Among several interpretations [8], a correlation coefficient can be thought of as the cosine of the angle (θ) between any two vectors $\boldsymbol{Y}$ and $\boldsymbol{X}$ containing a certain number of observations.[1] If the vectors are mean-centered ($\overline{Y}$, $\overline{X}$), $\cos\theta$ yields Pearson's $\boldsymbol{r}$.

$$r = \frac{(\mathbf{Y} - \overline{Y}\mathbf{1})^T(\mathbf{X} - \overline{X}\mathbf{1})}{\|\mathbf{Y} - \overline{Y}\mathbf{1}\|\|\mathbf{X} - \overline{X}\mathbf{1}\|} = \cos\theta \tag{1}$$

If the vectors are noncentral, $\cos\theta$ represents another correlation coefficient (denoted here as ρ) which has found application in drug chemical profiling [9–13] and document ink age estimation [14].

$$\rho = \frac{\boldsymbol{Y}^T\boldsymbol{X}}{\|\mathbf{Y}\|\|\mathbf{X}\|} = \cos\theta \tag{2}$$

Their squared versions (r^2, ρ^2) are frequently used when there is no need to distinguish between positive and negative correlations and they are more naturally related to sums of squares in analysis of variance tests [15]. Traditionally, vector $\boldsymbol{Y}$ represents a (so-called) dependent random

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[1] Capital bold face is used to represent column vectors. The notation $\mathbf{1}$ indicates an all-ones vector of the proper dimension and $\|\ \|$ represents vector Euclidean norm.

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variable whereas X denotes a linear combination of a certain number M of (so-called) independent variates which are maximally correlated using a common number N of experimental measurements [24].

A different approach is used in this work. Y and X are understood as two independent random vectors each one consisting respectively of a certain number N and M of random variables (dimensions). According to this new interpretation, any random vector of a given dimension is seen as a data set having that number of entries (size) and these terms are used interchangeably.

2. Materials and methods

2.1. Experimental procedure

2.1.1. Chemicals and reagents

Cocaine hydrochloride standards used in calibration were purchased from United States Pharmacopeia (USP), chromatographic grade chloroform 99.9% from Fermont, tetracosane (internal standard) from Sigma-Aldrich and sodium sulfate 99.5% from Riedel de Haën.

2.1.2. Samples

The dataset was composed of 523 cocaine hydrochloride samples from 449 seizures performed by the National Police of Peru (PNP) in different regions of Peru during the period between September 2021 and October 2022 (Fig. 1). All samples were analyzed in the laboratory of the PNP Center for Research and Strategic Studies against Illicit Drug Trafficking (CENIEETID).

2.1.3. GC-FID measurements

Ten milligrams of each cocaine sample were directly weighed into a graduated flask, mixed and sonicated with 10.00 mL of a 500 mg/L tetracosane solution in chloroform. Gas chromatography with Flame Ionization Detection (GC-FID) was performed with a Trace 1300 (Thermo Fisher Scientific) equipped with a TG-5MS capillary column (30 m × 250 μm x 0.25 μm). Helium was used as carrier gas; the injection port was maintained at a temperature of 280 °C and a split ratio of 20:1 was used. Column oven temperature program: initial temperature of 180 °C for 3 min; followed by linear 10 °C/min increments up to 280 °C and then maintained for 1 min [26]. Total chromatographic run time: 14 min.

2.1.4. HS-GC-MS measurements

Thirty milligrams of cocaine hydrochloride were directly weighed into a 20 mL headspace vial and then mixed with 10.00 mL of a 22 % m/V sodium sulfate aqueous solution. The vial was sealed and sonicated until the sample was properly dissolved. Residual solvent analysis [16] by Headspace Gas Chromatography coupled with Mass Spectrometry (HS-GC-MS) was performed with a 7697A headspace sampler (Agilent Technologies). Samples were heated at 85 °C for 14 min, mixed continuously, equilibrated for 0.05 min, pressurized for 0.20 min and the sample loop temperature was fixed at 175 °C. Gas chromatography was performed in an Agilent 8890 GC device equipped with a HP-5MS capillary column (30 m × 250 μm x 0.25 μm). Helium was used as carrier gas; the injection port was maintained at a temperature of 250 °C and a split ratio of 20:1 was used. Column oven temperature program: initial temperature of 36 °C for 13 min; followed by linear 6 °C/min increments up to 150 °C and 15 °C/min increments up to 200 °C. Total chromatographic run time: 35.33 min. Detection was performed with a quadrupole mass spectrometer (Agilent 5977B GC/MSD) and m/z range of 29–125 Da. Compound identification was performed with NIST17 database (NIST/EPA/NIH Mass Spectral Search Program) [17].

2.2. Data analysis and software

Data acquisition and automatic integration was accomplished with the MSD ChemStation E.02.021431 software (Agilent Technologies). The data matrix was exported to Excel® for statistical analysis (Table 1).

3. Theory

3.1. Correlation coefficient distribution – fixed dimensions

Consider two random vectors Y and X of respective fixed dimensions (number of random variables) N and M ($N \geq M$) representing two independent data sets and their sizes. Typical examples of fixed and well-defined dimensions are the total number of variable levels used in regression analysis and experimental design. The degree of proximity between both sets, can be established by the angle (θ) between them. If

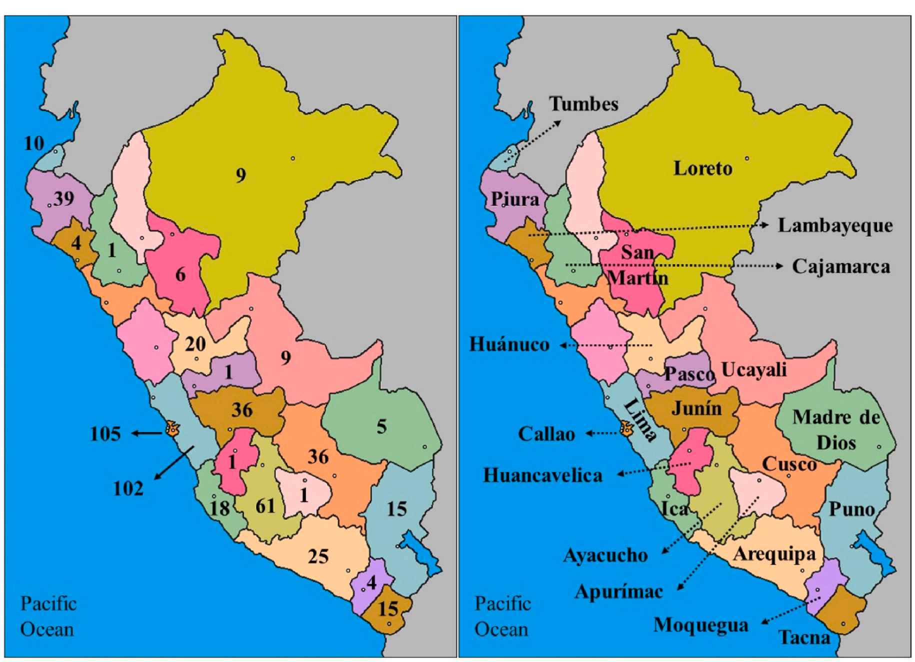

Fig. 1. Number of cocaine samples (left) according to Peru Departments (right).

Table 1
Excel® statistical functions used in this work.

Name	Mathematical Symbol	Excel® Function
Gamma function	$\Gamma(n)$	GAMMA(n)
Beta function[a]	$B(n,m)$	$\frac{\text{GAMMA}(n) \times \text{GAMMA}(m)}{\text{GAMMA}(n+m)}$
Cumulative Beta-distribution	$I_x(n,m)$	BETA.DIST(x,n,m,TRUE)
Inverse of the cumulative Beta-distribution	$u_{crit}(n,m)$	BETA.INV(Probability,n,m)

[a] The use of the Excel® function GAMMALN is required when high values of n and m are involved.

the data are independent, all possible angles can occur without restrictions and vectors may be regarded as points on the surface of an N-dimensional sphere of unit radius as θ is scale-independent (Fig. 2).

In the case of mean-centering, dimensions N and M need to be adjusted since one degree of freedom is lost for each vector. Additionally, if the vectors have the same dimension, an additional degree of freedom is lost due to the orthogonal projection of $\boldsymbol{Y}$ over $\boldsymbol{X}$. Therefore, the number of degrees of freedom (corrected dimensions) for each vector is calculated as follows.

$$\begin{cases} n = N - \delta_c \\ m = M - \delta_c - \delta_{N,M} \end{cases} \tag{3}$$

Where

$$\delta_c = \begin{cases} 1, \text{mean centering} \\ 0, \text{otherwise} \end{cases} \tag{4}$$

$$\delta_{N,M} = \begin{cases} 1, N = M \\ 0, \text{otherwise} \end{cases} \tag{5}$$

This approach allows handling both Pearson and noncentral ρ correlation coefficients within the same framework. Each vector contributes with a point in the $n-1$ dimensional surface S of the sphere. Hence, the probability of occurrence of an infinitesimal angle $d\theta$ between two vectors is proportional to its differential surface element dS which in

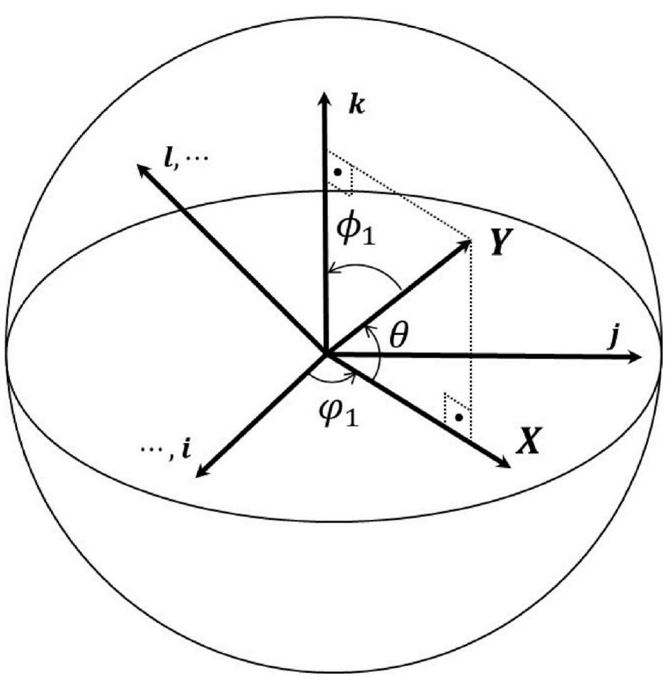

Fig. 2. Schematic representation of an n-dimensional sphere. Where: $\cdots, \boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k}, \boldsymbol{l}, \ldots$ = unit coordinate vectors.

spherical coordinates [18] can be calculated by the following equation.

$$dS = (\sin\theta)^{n-m-1} |\cos\theta|^{m-1}\, d\theta\, d\phi_{n-m-1}\, d\varphi_{m-1} \prod_{i=1}^{n-m-2} \sin\phi_i d\phi_i \prod_{j=1}^{m-2} \sin\varphi_j d\varphi_j \tag{6}$$

Where the ϕ_i 's represent the angles with respect to the exceeding $n-m-1$ dimensions of $\boldsymbol{Y}$ with respect to $\boldsymbol{X}$ ($0 \le \phi_i \le \pi$, $0 \le \phi_{n-m-1} < 2\pi$), the φ_j 's are the angles with respect to the common $m-1$ dimensions of $\boldsymbol{Y}$ and $\boldsymbol{X}$ ($0 \le \varphi_j \le \pi$) and $|\cos\theta|$ is used here to avoid negative areas. Thus, the cumulative distribution function (cdf) P_θ is obtained by the appropriate integration and normalization of equation (6) with respect to all angles involved.

$$P_\theta(n,m,\theta_f) = \text{Probability}(\Theta \le \theta) = \frac{\int_0^\theta (\sin\Theta)^{n-m-1}|\cos\Theta|^{m-1}\, d\Theta}{\int_0^{\theta_f} (\sin\Theta)^{n-m-1}|\cos\Theta|^{m-1}\, d\Theta} \tag{7}$$

Where the final integration angle θ_f may assume two different values.

$$\theta_f = \begin{cases} \pi/2, \text{only positive correlations are possible} \\ \pi, \text{negative correlations are also possible} \end{cases} \tag{8}$$

The case in which only positive (strictly speaking ≥ 0) correlations are possible is applicable to raw (noncentral) vectors having the same direction. An example of this situation is using noncentral ρ in the comparison of nonnegative chromatographic peaks or spectral bands. It is important to mention that a mean-centered coefficient such as Pearson's r may still assume negative values even if the vectors possess the same direction.

Equation (7) is an extension of the work of Pecka and Ponec [7] which was restrained to nonnegative correlations and used the traditional interpretation of dimensions n, m (see Introduction). Although it can be solved recursively in analytical form, no general closed-form solution is available. Instead, it is more advantageous to use its symmetry around $\theta = \pi/2$ and compute it with the help of the Beta function $B(m,n)$ [19].

$$B(n,m) = \int_0^1 u^{n-1}(1-u)^{m-1} du = \frac{\Gamma(n)\Gamma(m)}{\Gamma(n+m)}, n > 0, m > 0 \tag{9}$$

Where Γ stands for the gamma function [19].

In which case

$$P_\theta(n,m,\theta_f) = \frac{\int_0^\theta (\sin\Theta)^{n-m-1}|\cos\Theta|^{m-1}\, d\Theta}{C(n,m,\theta_f)}, 0 \le \theta \le \theta_f \tag{10}$$

$$C(n,m,\theta_f) = \frac{\theta_f}{\pi} B\left(\frac{n-m}{2}, \frac{m}{2}\right) \tag{11}$$

The probability density function (pdf) p_θ is obtained by taking the derivative of equation (10) with respect to θ.

$$p_\theta(n,m,\theta_f) = \frac{(\sin\theta)^{n-m-1}|\cos\theta|^{m-1}}{C(n,m,\theta_f)}, 0 \le \theta \le \theta_f \tag{12}$$

If necessary, the pdf of r^2 may be found from equation (12) by means of the transformation [20].

$$p_{r^2}(n,m) = p_\theta(n,m,\pi)\left|\frac{d\theta}{dr^2}\right| = \frac{1}{2}\beta_{r^2}\left(\frac{m}{2}, \frac{n-m}{2}\right) \tag{13}$$

Where β stands for the Beta distribution [20]. Except for the factor of $1/2$ and a different interpretation of dimensions n and m (see Introduction), equation (13) coincides with Fisher's multiple correlation coefficient distribution for normally distributed independent variates [3],[24]. This difference in normalizing factor occurs because Fisher's r^2 distribution does not distinguish between positive and negative correlations which is necessary in this work. Normalization discrepancy does not occur for the pdf of r. For instance, Fisher's bivariate case may be directly obtained from equation (12) by letting $m = 1$ [21].

$$p_r(n,1) = p_\theta(n,1,\pi)\left|\frac{d\theta}{dr}\right| = \frac{(1-r^2)^{\frac{n-3}{2}}}{B\left(\frac{1}{2},\frac{n-1}{2}\right)} \tag{14}$$

An example of $p_\theta(n,m,\theta_f)$ can be seen in Fig. 3. It is symmetric around $\pi/2$ and it has two variable maxima at $\theta_{max} = \tan^{-1}\left(\sqrt{(n-m-1)/(m-1)}\right)$ and $\pi - \theta_{max}$ (Supplementary Material). This distribution allows one to evaluate if an angle (θ) corresponding to a certain correlation coefficient is not the result of pure chance by checking it against a critical value.

$$\begin{cases} \theta \le \theta_{crit} \Rightarrow \text{significant correlation} \\ \theta > \theta_{crit} \Rightarrow \text{nonsignificant correlation} \end{cases} \tag{15}$$

This test may be framed in terms of correlation coefficients according to

$$\begin{cases} r \text{ or } \rho \ge \cos\theta_{crit} \Rightarrow \text{significant correlation} \\ r \text{ or } \rho < \cos\theta_{crit} \Rightarrow \text{nonsignificant correlation} \end{cases} \tag{16}$$

The critical angle is calculated by the condition (Fig. 3)

$$\int_0^{\theta_{crit}} p_\theta(n,m,\theta_f)\, d\theta = \alpha \tag{17}$$

From which the correlation coefficient critical value may be explicitly calculated (Appendix A).

$$r_{crit}(n,m,\theta_f) \text{ or } \rho_{crit}(n,m,\theta_f) = \sqrt{1-u_{crit}(n,m,\theta_f)} \tag{18}$$

$$u_{crit}(n,m,\theta_f) = \text{BETA.INV}\left(\frac{2\theta_f}{\pi}\alpha, \frac{n-m}{2}, \frac{m}{2}\right) \tag{19}$$

Where BETA.INV is an Excel® statistical function, which returns the inverse of the cumulative Beta distribution (Table 1). Notice how the term $2\theta_f/\pi$ automatically adjusts for the required significance level.

3.2. Correlation coefficient distribution – variable dimensions

In real-world matching, such as chromatogram or spectral comparisons, data sizes are rarely fixed as the number of peaks and chemical compounds are sample dependent and affected by experimental conditions (analytical method, detection limits, background levels, equipment fluctuations, minor compounds, contaminants etc.). Consequently, the correlation coefficient distribution must account for all possible combinations between a given vector $\boldsymbol{Y}$ having n degrees of freedom (leading dimension) and any another vector $\boldsymbol{X}$ holding at most n degrees of freedom ($m \le n$). According to equation (3), m may assume values ranging from 1 up to $n-1$. Therefore, equation (12) must be properly modified and normalized according to the next equation.

$$p_\theta(n,\theta_f) = \frac{\sum_{k=0}^{n-2}\binom{n-2}{k}(\sin\theta)^{n-2-k}|\cos\theta|^k}{\int_0^{\theta_f}\sum_{k=0}^{n-2}\binom{n-2}{k}(\sin\theta)^{n-2-k}|\cos\theta|^k d\theta}, 0 \le \theta \le \theta_f \tag{20}$$

Which, as a result of the binomial expansion formula [19], reduces to

$$p_\theta(n,\theta_f) = \frac{(\sin\theta + |\cos\theta|)^{n-2}}{\int_0^{\theta_f}(\sin\theta+|\cos\theta|)^{n-2}d\theta}, 0 \le \theta \le \theta_f, n \ge 2 \tag{21}$$

Equation (21) is consistent with the fact that at least two degrees of freedom are required in order to avoid the trivial case of colinear vectors. It may be alternatively expressed by the following equations (Appendix B).

$$p_\theta(n,\theta_f) = \begin{cases} \dfrac{\sin^{n-2}\left(\theta+\frac{\pi}{4}\right)}{D(n,\theta_f)}, 0 \le \theta \le \dfrac{\pi}{2} \\ \dfrac{\sin^{n-2}\left(\theta-\frac{\pi}{4}\right)}{D(n,\theta_f)}, \dfrac{\pi}{2} < \theta \le \pi \end{cases} \tag{22}$$

$$D(n,\theta_f) = \frac{2\theta_f}{\pi}B\left(\frac{n-1}{2},\frac{1}{2}\right)\left[1 - I_{1/2}\left(\frac{n-1}{2},\frac{1}{2}\right)\right] \tag{23}$$

If required, similar transformations to the ones in (13) and (14) may be applied to equation (22) in order obtain the explicit pdfs of r, r^2, ρ and ρ^2 (Appendix C). An example of $p_\theta(n,\theta_f)$ is shown in Fig. 4. This distribution is symmetric around $\theta = \{\pi/4, \pi/2, 3\pi/4\}$ and it has two fixed maxima at $\theta_{max} = \pi/4$ and $\pi - \theta_{max} = 3\pi/4$. It can be shown [23] (Supplementary Material) that these maxima also represent the limit critical angles as the leading dimension tends to very large values ($n \to \infty$). Fig. 5 illustrates this behavior for the function $\sin^{n-2}(\theta+\pi/4)$ which is the bounded version of $p_\theta(n,\theta_f)$ in the interval $0 \le \theta \le \pi/2$.

Under the reasonable assumption that the required significance level (α) will never be higher than 50% for $\theta_f = \pi/2$ or 25% for $\theta_f = \pi$, it follows that $0 \le \theta_{crit} \le \pi/4$. Hence, in the case of variable dimension data comparisons, a minimum correlation coefficient r or ρ match of approximately 70% (or equivalently r^2 or ρ^2 = 50%) is required to achieve proper statistical significance.

$$\min_{n\to\infty}\left[r_{crit}(n,\theta_f) \text{ or } \rho_{crit}(n,\theta_f)\right] = \cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2} \cong 0.707 \tag{24}$$

$$\min_{n\to\infty}\left[r^2_{crit}(n,\theta_f) \text{ or } \rho^2_{crit}(n,\theta_f)\right] = \cos^2\left(\frac{\pi}{4}\right) = 0.5 \tag{25}$$

As in the previous case, the critical angle is calculated by the

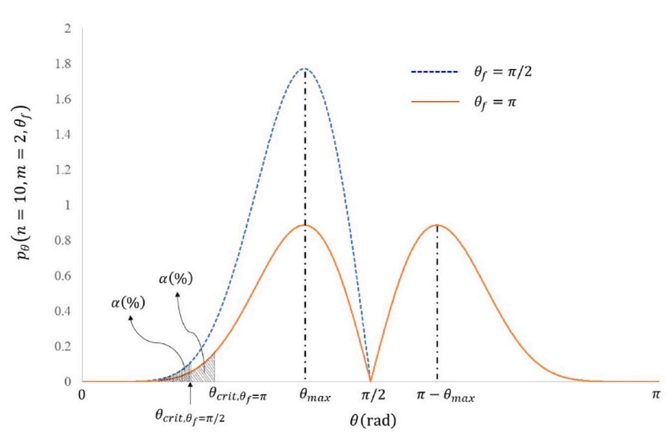

Fig. 3. Illustrative example of p_θ (n,m,θ_f) – fixed size comparisons.

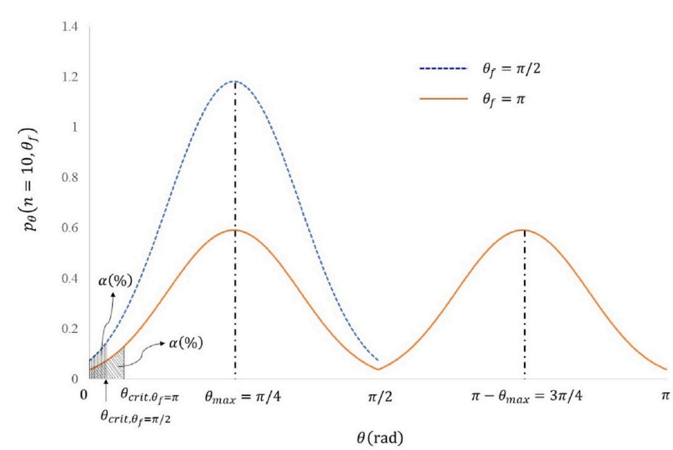

Fig. 4. Illustrative example of p_θ (n,θ_f) – variable size comparisons.

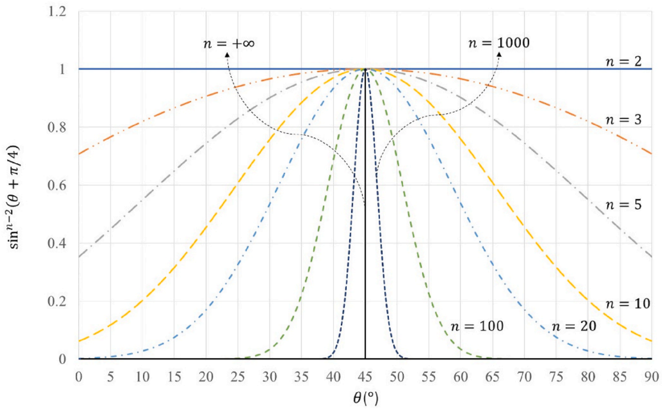

Fig. 5. Limit behavior of $\sin^{n-2}(\theta + \pi/4)$ as $n \to \infty$.

condition (Fig. 4)

$$\int_0^{\theta_{crit}} p_\theta(n, \theta_f)\, d\theta = \alpha \tag{26}$$

From which the correlation coefficient critical value may be explicitly calculated (Appendix D).

$$r_{crit}(n, \theta_f) \text{ or } \rho_{crit}(n, \theta_f) = \sqrt{\frac{1 + \sqrt{1 - \left(1 - 2u_{crit}(n, \theta_f)\right)^2}}{2}} \tag{27}$$

$$u_{crit}(n, \theta_f) = \text{BETA.INV}\left(\text{Probability}, \frac{n-1}{2}, \frac{1}{2}\right) \tag{28}$$

$$\text{Probability} = \frac{4\theta_f}{\pi}\alpha\left[1 - I_{1/2}\left(\frac{n-1}{2}, \frac{1}{2}\right)\right] + I_{1/2}\left(\frac{n-1}{2}, \frac{1}{2}\right) \tag{29}$$

$$I_{1/2}\left(\frac{n-1}{2}, \frac{1}{2}\right) = \text{BETA.DIST}\left(\frac{1}{2}, \frac{n-1}{2}, \frac{1}{2}, \text{TRUE}\right) \tag{30}$$

Where BETA.DIST and BETA.INV are Excel® statistical functions which return respectively the cumulative Beta distribution and its inverse (Table 1). Correlation coefficient critical values calculated according to equations (27)–(30) for $\alpha = 1\%$, 5% are shown in Table 2.

4. Results and discussion

4.1. Fixed dimensions

4.1.1. "Number of nines" in calibration curve fitting

Table 3 exhibits the calibration data used in the quantification of cocaine by GC-FID. Cocaine concentration ranged from 7.8 to 2000 mg/L over nine calibration levels with three replicates at each level. Fig. 6 shows the linear least squares regression performed with the ratios of cocaine to internal standard concentrations [C]/[IS]x100 as the dependent variable ($\boldsymbol{X}$) and the cocaine to internal standard chromatographic areas (A_C/A_{IS})x100 as the independent variable ($\boldsymbol{Y}$). Regression quality is commonly expressed by the coefficient of multiple determination [15], typically defined as the square of the Pearson correlation coefficient between the predicted values ($\widehat{\boldsymbol{Y}}$) and the replicate averages at each regression level ($\boldsymbol{Y}_{avg}$).

$$r^2_{calc} = \frac{\sum_{i=1}^{9}\left(\widehat{Y}_i - \overline{Y}\right)^2}{\sum_{i=1}^{9}\left(Y_{avg,i} - \overline{Y}\right)^2} = 0.99996$$

Where

$$\overline{Y} = \frac{\sum_{i=1}^{9}\sum_{j=1}^{3} Y_{ij}}{3 \times 9} = 52.359$$

Although r^2_{calc} is extremely high and very close to one, the question remains as to what would be the critical value for a regression to be considered significant in general. This can be attained within the framework of this work by considering $\boldsymbol{Y}_{avg}$ and $\widehat{\boldsymbol{Y}}$ as two random vectors

Table 2
Correlation coefficient critical values – variable dimensions.

n	Only Positive Correlations $\theta_f = \pi/2$				Possible Negative Correlations $\theta_f = \pi$			
	α = 1%		α = 5%		α = 1%		α = 5%	
	ρ_{crit}	ρ^2_{crit}	ρ_{crit}	ρ^2_{crit}	r_{crit}	r^2_{crit}	r_{crit}	r^2_{crit}
2	0.9999	0.9998	0.9969	0.9938	0.9995	0.9990	0.9877	0.9755
3	0.9998	0.9996	0.9954	0.9909	0.9992	0.9985	0.9831	0.9665
4	0.9997	0.9994	0.9934	0.9868	0.9988	0.9976	0.9775	0.9555
5	0.9995	0.9990	0.9906	0.9813	0.9981	0.9963	0.9710	0.9429
6	0.9992	0.9984	0.9871	0.9743	0.9972	0.9943	0.9639	0.9292
7	0.9987	0.9974	0.9829	0.9660	0.9958	0.9916	0.9565	0.9149
8	0.9980	0.9960	0.9780	0.9566	0.9939	0.9879	0.9490	0.9005
9	0.9970	0.9940	0.9728	0.9463	0.9916	0.9832	0.9415	0.8865
10	0.9956	0.9913	0.9673	0.9356	0.9887	0.9776	0.9344	0.8730
11	0.9939	0.9878	0.9617	0.9248	0.9854	0.9711	0.9275	0.8603
12	0.9917	0.9835	0.9561	0.9140	0.9818	0.9639	0.9211	0.8484
13	0.9892	0.9785	0.9506	0.9035	0.9779	0.9563	0.9150	0.8373
14	0.9863	0.9729	0.9452	0.8934	0.9739	0.9484	0.9094	0.8269
15	0.9833	0.9668	0.9401	0.8838	0.9697	0.9404	0.9041	0.8173
16	0.9800	0.9605	0.9352	0.8746	0.9656	0.9324	0.8991	0.8084
17	0.9767	0.9540	0.9305	0.8659	0.9616	0.9246	0.8945	0.8002
18	0.9733	0.9474	0.9261	0.8576	0.9576	0.9169	0.8902	0.7925
19	0.9700	0.9408	0.9219	0.8499	0.9537	0.9095	0.8862	0.7853
20	0.9666	0.9343	0.9179	0.8425	0.9499	0.9023	0.8824	0.7786
30	0.9371	0.8781	0.8871	0.7869	0.9189	0.8444	0.8543	0.7299
50	0.8985	0.8073	0.8524	0.7265	0.8811	0.7763	0.8243	0.6794
100	0.8514	0.7249	0.8139	0.6624	0.8369	0.7004	0.7922	0.6275
200	0.8135	0.6618	0.7846	0.6156	0.8022	0.6435	0.7684	0.5904
500	0.7768	0.6033	0.7572	0.5734	0.7690	0.5914	0.7465	0.5573
+ ∞	0.7071	0.5000	0.7071	0.5000	0.7071	0.5000	0.7071	0.5000

Table 3
Calibration data for cocaine quantification by GC-FID

Calibration Level	$\boldsymbol{X}$	Replicates			$\boldsymbol{Y}_{avg}$	$\widehat{Y}$
		$\boldsymbol{Y}_1$	$\boldsymbol{Y}_2$	$\boldsymbol{Y}_3$		
1	1.562	0.852	0.633	0.680	0.722	0.407
2	3.125	1.449	1.470	1.585	1.501	1.338
3	6.249	3.361	3.272	3.087	3.240	3.201
4	12.499	7.183	6.728	6.367	6.759	6.927
5	24.998	15.705	14.519	14.122	14.782	14.378
6	49.995	30.172	29.259	29.043	29.491	29.280
7	99.991	59.355	58.337	57.949	58.547	59.085
8	199.981	118.593	117.041	117.383	117.673	118.696
9	399.962	237.650	238.425	239.463	238.513	237.916

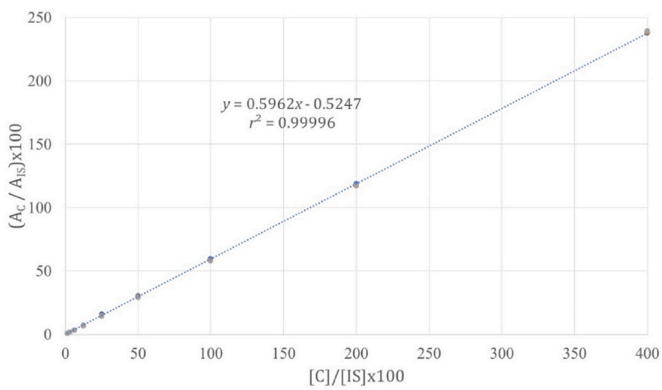

Fig. 6. Calibration curve for cocaine quantification by GC-FID.

with the same dimension L and equal to the number of calibration levels. Following equation (3), $\boldsymbol{Y}_{avg}$ can be assigned $n = L - 1$ and $\widehat{\boldsymbol{Y}}$ $m = L - 2$ degrees of freedom. The application of equations (18) and (19) results

$$r^2_{crit}(L-1, L-2, \pi) = 1 - \text{BETA.INV}\left(0.10, \frac{1}{2}, \frac{L-2}{2}\right) \tag{31}$$

Table 4 exhibits the coefficient of multiple determination critical values calculated according to equation (31) as a function of the number of calibration levels L at the confidence level $\alpha = 5\%$. For the case at hand ($L = 9$), $r^2_{calc} = 0.99996 > r^2_{crit}(8, 7, \pi) = 0.9976$ confirmed the good quality of fit. As regression also passed both residuals plot and lack of fit tests (results not shown) the calibration curve has been used to quantify all cocaine samples.

It is important to notice that $r^2_{crit}(L-1, L-2, \pi)$ provides a statistical basis for the required coefficient of multiple determination ("number of nines") in general mathematical model fitting. It does not depend either on the type (linear, nonlinear) or the number of model parameters and may be used as a calibration curve acceptance criterion in validation protocols.

Table 4
Coefficient of multiple determination critical values versus number of calibration levels (L), $\alpha = 5\%$.

L	$r^2_{crit}(L\text{-}1, L\text{-}2, \pi)$ [a]	L	$r^2_{crit}(L\text{-}1, L\text{-}2, \pi)$ [a]
1	n/a	11	0.9981
2	n/a	12	0.9983
3	0.9755	13	0.9985
4	0.9900	14	0.9986
5	0.9938	15	0.9987
6	0.9955	16	0.9988
7	0.9965	17	0.9989
8	0.9971	18	0.9990
9	0.9976	19	0.9990
10	0.9979	20	0.9991

[a] Calculated according to equation (31).

4.2. *Variable dimensions*

4.2.1. *Cocaine chemical profiling*

The dataset was composed of 523 cocaine hydrochloride samples from 449 seizures performed by PNP in different Departments of Peru (Fig. 1). All samples have been analyzed for cocaine purity (GC-FID) and residual solvents (HS-GC-MS).

A high number of solvents in a sample potentially provides more useful information regarding cocaine handling and processing as well as increases the statistical relevance of a possible correlation. Therefore, the thirteen samples in the set containing ten or more residual solvent peaks (Table 5) were tested as candidate matches.

Each HS-GC-MS chromatogram was treated as a random vector whose dimension varies with the number of numerically integrated peaks (N) of each chemical compound. According to standard validation procedures [22], chromatographic peaks are considered relevant for component identification if their signal-to-noise ratio (S/N) is ≥ 3 and for component quantification if S/N ≥ 10. Numerical simulations have shown that these conditions have been achieved when the ChemStation® Integrator sensitivity parameter Initial Threshold (IT) was set to IT = 15.5 (S/N $\geq$ 3) and IT = 18 (S/N $\geq$ 10). The effect of minimum S/N on the number of compounds and sample correlation is seen in Table 6 for a candidate match between samples C1063 and C1071. As minimum S/N increased from 3 to 10, the total number of integrated peaks (N) decreased from 44 to 19 but the calculated correlation coefficients did not change significantly staying well above their critical values. This result has been confirmed for other samples in the set. Fig. 7 shows a comparison between samples C1063 and C1071 in terms of relative chromatographic peak areas normalized to the most abundant

Table 5
Samples containing ten or more peaks ($N \geq 10$).

Item	Sample	N	Seizure N°	Department	Logo	Cocaine (%)
1	C357	28	118	Callao	PML1	82.9
2	C404	22	134	Arequipa	n/a	89.2
3	C347	20	114	Callao	n/a	82.0
4	C186	19	59	Callao	n/a	78.6
5	C1063	19	433	Tumbes	n/a	73.7
6	C1071	19	435	Tumbes	8F	80.3
7	C42	17	16	Madre de Dios	G	85.0
8	C38	12	14	Ayacucho	DELFIN	86.5
9	C70	12	30	Huánuco	MERCEDES BENZ	81.3
10	C941	12	376	Loreto	AUDI	86.8
11	C873	11	354	Tumbes	n/a	65.7
12	C803	10	325	Piura	MR	94.3
13	C939	10	375	Loreto	PEAKY	84.3

Table 6
Integration parameters for candidate match between samples C1063 and C1071.

Parameter	$(S/N)_{min}$	N	$r_{crit}(n,\pi)$[a]	r_{calc}	$\rho^2_{crit}(n,\pi/2)$[a]	ρ^2_{calc}
IT = 15.5	3	44	0.8325	0.9712	0.7406	0.9539
IT = 18	10	19	0.8902	0.9509	0.8499	0.9432
TIC	n/a	n/a	n/a	0.9674	n/a	0.9493

[a] Calculated according to equations (27)–(30) for $\alpha = 5\%$.

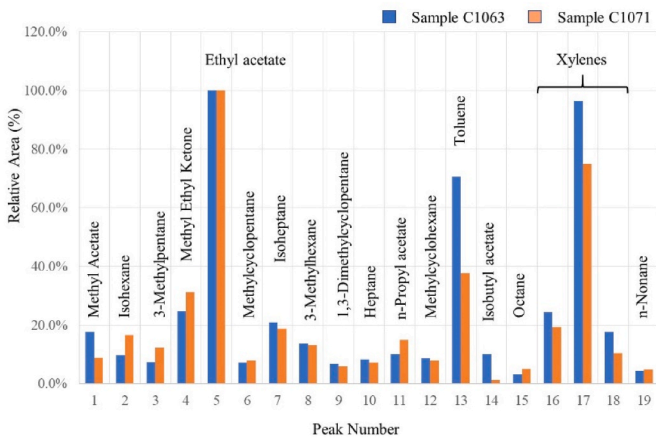

Fig. 7. Relative chromatographic peak area comparison between samples C1063 and C1071, IT = 18.

peak (ethyl acetate) for IT = 18.

The procedure of lining up components and their respective chromatographic areas for each possible match is not efficient for practical routine lab work since it becomes progressively cumbersome and time consuming as the number of components and samples increases. Thus, the Total Ion Count (TIC) chromatogram of an entire run was evaluated as an alternative way to more efficiently calculate pairwise correlation coefficients. Table 6 shows the results for candidate match C1063 – C1071 where the obtained correlation coefficients ($r_{calc} = 0.9674$, $\rho^2_{calc} = 0.9493$) fell reasonably well between the corresponding values for $(S/N)_{min} = 3$ and 10 and were considered as acceptable. Similar outcomes have been verified for other samples in the set such that the TIC chromatogram of each run was used for correlation coefficient calculations and integration parameter IT = 18 was employed in establishing the number of peaks (N) in all further comparisons.

Noncentral cosine squared (ρ^2) and Pearson coefficient (r) correlation matrices are shown respectively in Table 7 and Table 8. Critical values were calculated according to equations (27)–(30). Matches based on residual solvent analysis are marked in underlined bold face. No significant differences have been observed between choosing either r or ρ^2 as correlation coefficients. Added evidence (Table 5) can be used to strengthen or weaken suggested linkages. For instance, it might be considered of relevance that samples C1063 and C1071 originated from very close seizures (433 and 435) in the same Department (Tumbes). It could also be important that samples C939 and C941 were seized sequentially (seizure numbers 375 and 376) in Loreto, present similar cocaine contents (84.3% and 86.8%) and both exhibit logos (Fig. 8). Such candidate links indicate similar crystallization/purification procedures and may be further validated by the presence/absence of relevant peaks, additional chemical analysis and further police information. TIC chromatogram raw data for the thirteen samples in the set with $N \geq 10$ can be found in the Supplementary Material.

5. Conclusions

This paper has presented a novel and comprehensive statistical framework in order to rigorously derive the correlation coefficient distribution between two independent chemical data sets. A systematic derivation in the angle domain set the theoretical basis for fixed size comparisons and led to the generalization into the new and experimentally relevant case of variable dimensions. As a result, equations for the correlation coefficient distributions and their critical values have been explicitly obtained and expressed in terms of readily implementable standard Excel® statistical functions.

Both mean-centered (r, r^2) and noncentral (ρ, ρ^2) correlation coefficients have been included as the number of degrees of freedom has been properly accounted for. The introduction of normalization parameter θ_f allowed the distinction between only positive and possible negative correlations and the unambiguous calculation of significance levels (α) in each case.

The application of the theory to regression curve evaluation resulted in an objective criterion ("number of nines") for general mathematical model fitting and calibration curve acceptance in validation protocols.

Table 7
Noncentral cosine squared correlation coefficient (ρ^2) matrix.[a].

Sample	C357	C404	C347	C186	C1063	C1071	C42	C38	C70	C941	C873	C803	C939
C357	1												
C404	0.2865	1											
C347	0.4277	**<u>0.9329</u>**	1										
C186	0.4889	0.8006	**<u>0.8537</u>**	1									
C1063	0.2990	**<u>0.9073</u>**	**<u>0.9016</u>**	**<u>0.8814</u>**	1								
C1071	0.2145	**<u>0.9284</u>**	**<u>0.8514</u>**	0.7912	**<u>0.9493</u>**	1							
C42	0.0532	0.1258	0.1059	0.1570	0.1132	0.1289	1						
C38	0.3171	0.5011	0.5359	0.5853	0.4823	0.4476	0.4483	1					
C70	0.3460	**<u>0.8416</u>**	**<u>0.8471</u>**	0.6479	0.7462	0.7762	0.0890	0.4596	1				
C941	0.6702	0.1115	0.1433	0.1994	0.0962	0.0964	0.0339	0.1070	0.1564	1			
C873	0.4960	0.4954	0.4606	0.4537	0.4162	0.4901	0.1024	0.2781	0.5097	0.6909	1		
C803	0.0381	0.1411	0.0874	0.1415	0.1147	0.1159	0.6029	0.3394	0.0841	0.0231	0.0796	1	
C939	0.5409	0.0762	0.0931	0.1307	0.0578	0.0669	0.0224	0.0638	0.1229	**<u>0.9663</u>**	0.6405	0.0163	1
n	28	22	20	19	19	19	17	12	12	12	11	10	10
$\rho^2_{crit}(n,\pi/2)$[b]	0.7960	0.8290	0.8425	0.8499	0.8499	0.8499	0.8659	0.9140	0.9140	0.9140	0.9248	0.9356	0.9356

[a] TIC, IT = 18.
[b] Calculated according to equations (27)–(30) for $\alpha = 5\%$.

Table 8
Pearson correlation coefficient (r) matrix.[a]

Sample	C357	C404	C347	C186	C1063	C1071	C42	C38	C70	C941	C873	C803	C939
C357	1												
C404	0.4617	1											
C347	0.6037	**0.9582**	1										
C186	0.6600	0.8659	**0.9108**	1									
C1063	0.4750	**0.9400**	**0.9379**	**0.9241**	1								
C1071	0.3729	**0.9538**	**0.9046**	0.8545	**0.9674**	1							
C42	0.0617	0.1303	0.1097	0.1281	0.1029	0.1218	1						
C38	0.4973	0.6103	0.6657	0.6446	0.5873	0.5358	0.4675	1					
C70	0.5371	**0.9096**	**0.9108**	0.7828	0.8458	0.8681	0.1313	0.6441	1				
C941	0.8020	0.2434	0.2997	0.3678	0.2149	0.2120	0.0340	0.1976	0.3327	1			
C873	0.6625	0.6211	0.5965	0.5606	0.5447	0.6110	0.0633	0.3105	0.6656	0.8290	1		
C803	0.0486	0.1997	0.1121	0.1563	0.1505	0.1436	0.6965	0.3952	0.1495	0.0223	0.0645	1	
C939	0.7211	0.2064	0.2427	0.3003	0.1649	0.1835	0.0361	0.1513	0.3024	**0.9842**	0.8153	0.0294	1
n	27	21	19	18	18	18	16	11	11	11	10	9	9
$r_{crit}(n,\pi)$[b]	0.8613	0.8788	0.8862	0.8902	0.8902	0.8902	0.8991	0.9275	0.9275	0.9275	0.9344	0.9415	0.9415

[a] TIC, IT = 18.
[b] Calculated according to equations (27)–(30) for $\alpha = 5\%$.

Fig. 8. Cocaine brick logos of samples C939 (top) and C941 (bottom).

In the case of variable size data comparisons, it has been demonstrated that a minimum correlation coefficient r or $\rho \approx 70\%$ (or equivalently r^2 or $\rho^2 = 50\%$) is required for statistical relevance. Drug profiling (purity and residual solvents) together with available additional information (logos, seizure site etc.) were used in the real case linkage of 532 cocaine samples seized by the National Police of Peru between the years of 2021 and 2022.

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CRediT authorship contribution statement

Jorge Jardim Zacca: Writing – review & editing. **Carmen P. Sandoval Jáuregui:** Resources, Project administration, Funding acquisition. **Flor M. Villafana Bardales:** Supervision, Resources, Project administration, Data curation. **Johanna Remuzgo Yabar:** Validation, Investigation, Data curation. **Julia E. Enciso Soria:** Supervision, Resources. **Vanessa D. Ayala Caro:** Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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Appendix E. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.chemolab.2024.105091.

Appendix A

A change of variables $u = \cos^2\theta$ in equation (17) results

$$I_{u_{crit}}\left(\frac{n-m}{2},\frac{m}{2}\right)=\frac{2\theta_f}{\pi}\alpha \tag{A.1}$$

$$u_{crit}=\sin^2\theta_{crit} \tag{A.2}$$

Where $I_x(n,m)$ means the cumulative beta distribution (also known as the Regularized Incomplete Beta Function [25]).

$$I_x(n,m)=\frac{\int_0^x u^{n-1}(1-u)^{m-1}du}{B(n,m)} \tag{A.3}$$

From (A. 2)

$$\cos\theta_{crit}=\sqrt{1-u_{crit}(n,m,\theta_f)} \tag{A.4}$$

Appendix B

Consider the trigonometric identities

$$\sin\left(\theta+\frac{\pi}{4}\right)=\sin\theta\cos\left(\frac{\pi}{4}\right)+\sin\left(\frac{\pi}{4}\right)\cos\theta=\frac{\sqrt{2}}{2}(\sin\theta+|\cos\theta|),0\le\theta\le\frac{\pi}{2} \tag{B.1}$$

and

$$\sin\left(\theta-\frac{\pi}{4}\right)=\sin\theta\cos\left(\frac{\pi}{4}\right)-\sin\left(\frac{\pi}{4}\right)\cos\theta=\frac{\sqrt{2}}{2}(\sin\theta+|\cos\theta|),\frac{\pi}{2}<\theta\le\pi \tag{B.2}$$

Equations (B. 1) and (B. 2) imply that $p_\theta(n,\theta_f)$ is symmetric at the points $\pi/4$, $\pi/2$, and $3\pi/4$. Using only the symmetry around $\theta=\pi/4$ the denominator of equation (21) can be expressed as

$$\int_0^{\theta_f}(\sin\theta+|\cos\theta|)^{n-2}d\theta=\frac{4\theta_f}{\pi}\left(\sqrt{2}\right)^{n-2}\int_0^{\frac{\pi}{4}}\sin^{n-2}\left(\theta+\frac{\pi}{4}\right)d\theta \tag{B.3}$$

A proper change of variable $u=\sin^2(\theta+\pi/4)$ results

$$\frac{4\theta_f}{\pi}\int_0^{\frac{\pi}{4}}\sin^{n-2}\left(\theta+\frac{\pi}{4}\right)d\theta=\frac{2\theta_f}{\pi}B\left(\frac{n-1}{2},\frac{1}{2}\right)\left[1-I_{1/2}\left(\frac{n-1}{2},\frac{1}{2}\right)\right] \tag{B.4}$$

Thus, equation (21) may be alternatively expressed as

$$p_\theta(n,\theta_f)=\begin{cases}\dfrac{\sin^{n-2}\left(\theta+\frac{\pi}{4}\right)}{D(n,\theta_f)}, & 0\le\theta\le\dfrac{\pi}{2}\\[2ex] \dfrac{\sin^{n-2}\left(\theta-\frac{\pi}{4}\right)}{D(n,\theta_f)}, & \dfrac{\pi}{2}<\theta\le\pi\end{cases} \tag{B.5}$$

$$D(n,\theta_f)=\frac{2\theta_f}{\pi}B\left(\frac{n-1}{2},\frac{1}{2}\right)\left[1-I_{1/2}\left(\frac{n-1}{2},\frac{1}{2}\right)\right] \tag{B.6}$$

Appendix C

The pdfs for all correlation coefficients (r, ρ, r^2 and ρ^2) may be obtained from $p_\theta(n,\theta_f)$ equation (22) by means of the following transformations [20]

$$p_r(n)=p_{\theta(r)}(n,\pi)\left|\frac{d\theta}{dr}\right|=\frac{\left(\frac{\sqrt{1-r^2}+|r|}{\sqrt{2}}\right)^{n-2}}{D(n,\pi)\sqrt{1-r^2}},-1\le r\le 1 \tag{C.1}$$

$$p_{r^2}(n)=p_r(n)\left|\frac{dr}{dr^2}\right|=\frac{\left(\frac{\sqrt{1-r^2}+|r|}{\sqrt{2}}\right)^{n-2}}{2D(n,\pi)|r|\sqrt{1-r^2}},0\le r^2\le 1 \tag{C.2}$$

$$p_\rho(n)=p_{\theta(\rho)}\left(n,\frac{\pi}{2}\right)\left|\frac{d\theta}{d\rho}\right|=\frac{\left(\frac{\sqrt{1-\rho^2}+\rho}{\sqrt{2}}\right)^{n-2}}{D\left(n,\frac{\pi}{2}\right)\sqrt{1-\rho^2}},0\le\rho\le 1 \tag{C.3}$$

$$p_{\rho^2}(n) = p_\rho(n)\left|\frac{d\rho}{d\rho^2}\right| = \frac{\left(\frac{\sqrt{1-\rho^2}+\rho}{\sqrt{2}}\right)^{n-2}}{2D\left(n,\frac{\pi}{2}\right)\rho\sqrt{1-\rho^2}}, 0 \le \rho^2 \le 1 \tag{C.4}$$

Notice how scaling factor $D(n,\theta_f)$ conveniently adjusts for correlation coefficient range differences. Distributions (C. 1) to (C. 4) present infinite jump discontinuities at the points $r^2 = \rho^2 = 1$ while $p_{r^2}(n)$ and $p_{\rho^2}(n)$ have an additional discontinuity at $r = \rho = 0$. On the other hand, $p_\theta(n,\theta_f)$ developed in this work is a relatively well-behaved function (Fig. 4) being more suitable to population modeling.

Appendix D

From equations (22), (23) and (26)

$$\int_0^{\theta_{crit}} \frac{\sin^{n-2}\left(\theta+\frac{\pi}{4}\right)}{B\left(\frac{n-1}{2},\frac{1}{2}\right)\left[1 - I_{1/2}\left(\frac{n-1}{2},\frac{1}{2}\right)\right]} d\theta = \frac{2\theta_f}{\pi}\alpha \tag{D.1}$$

Given that $0 \le \theta_{crit} \le \pi/4$, a change of variables $u = \sin^2(\theta+\pi/4)$ allows to rewrite equation (D. 1) as

$$\frac{I_{u_{crit}}\left(\frac{n-1}{2},\frac{1}{2}\right) - I_{1/2}\left(\frac{n-1}{2},\frac{1}{2}\right)}{1 - I_{1/2}\left(\frac{n-1}{2},\frac{1}{2}\right)} = \frac{4\theta_f}{\pi}\alpha \tag{D.2}$$

$$u_{crit} = \sin^2\left(\theta_{crit} + \frac{\pi}{4}\right) \tag{D.3}$$

From (D. 3)

$$\left(\cos^2\theta_{crit}\right)^2 - \cos^2\theta_{crit} + \left(\frac{1}{2} - u_{crit}\right)^2 = 0 \tag{D.4}$$

Since $0 \le \theta_{crit} \le \pi/4$ the only feasible solution to (D. 4) is

$$\cos\theta_{crit} = \sqrt{\frac{1+\sqrt{1-\left(1-2u_{crit}(n,\theta_f)\right)^2}}{2}} \tag{D.5}$$

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